Tests and Homeworks

Balázs Lengyel

Economic Complexity, Class 2 – Sept 23 2026

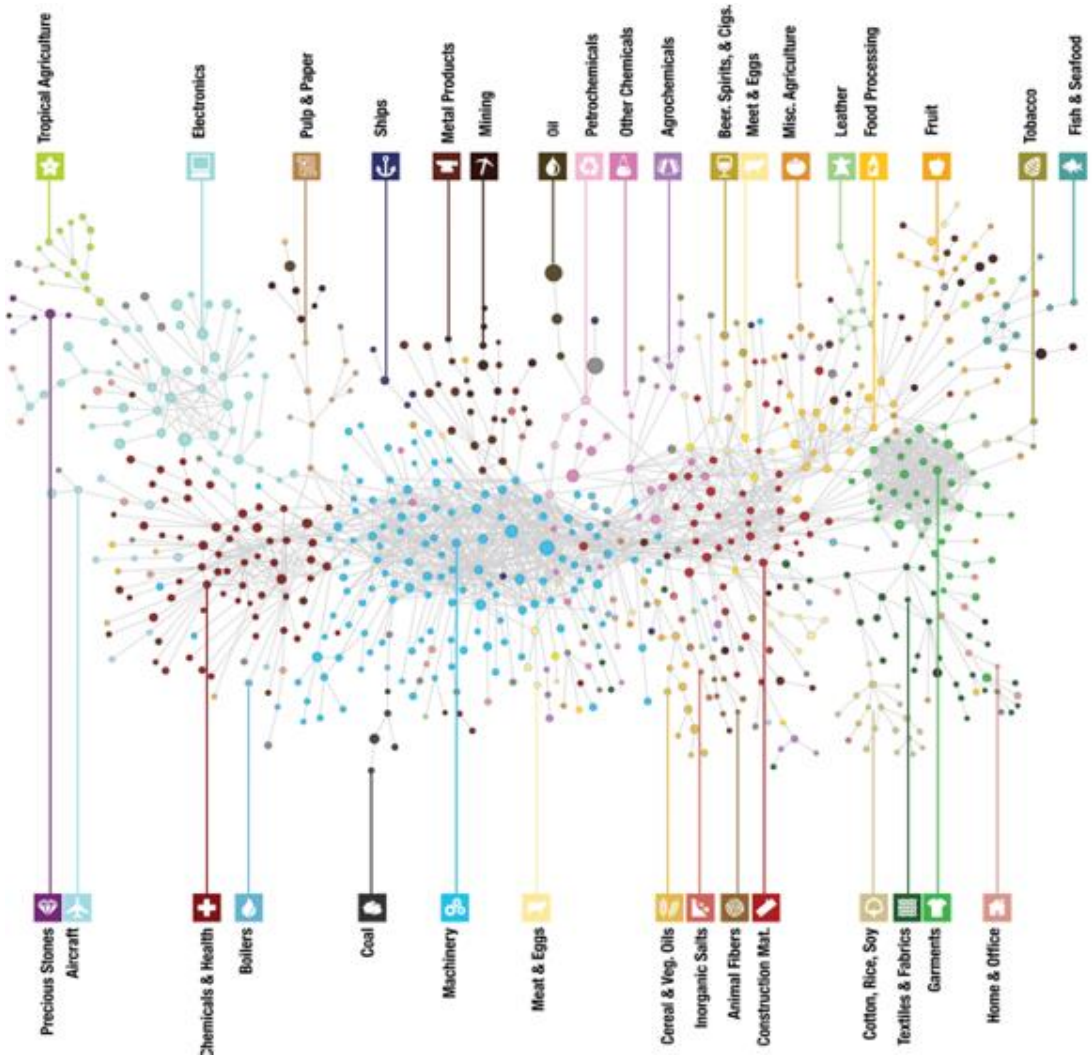
Social Data Science, Corvinus University of Budapest

Quiz 1 – Sept 23

1. What are the assumptions behind the product space? Describe at least 2. (2 points)

2. What are the empirical observations made with the use of product space regarding diversification? (1 point)

Observations



Assumptions

Statements that are not verified but the measurement and conclusion rely on them

- RCA can measure capabilities
- Products frequently exported together with RCA>1 share similar capabilities

HW1 for Sept 28

Interesting problem and data.

1. Identify a topic that is connected to the course and interests you. This can be important for a company, city, an industry, a country, for specific groups of people or the global humanity (eg. automation, inequality, international competition, climate change, populism etc.).
2. Search for journal articles that address this topic with interesting data. Choose one that you would like to show to the class.
3. Prepare 1 pptx slide that summarizes the paper. The presentation cannot be longer than 1 slide. Best slides typically include figures that can illustrate the story.
4. Be able to present the slide and talk about the research problem, question, the data, methods, and findings. Save the file: ”givenname_familyname_EC2026_HW1.pptx”.
5. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
6. Homeworks covering 1-2-3 can get 2 points. Presentation gives an extra 1 point.

Reading for – Sept 30

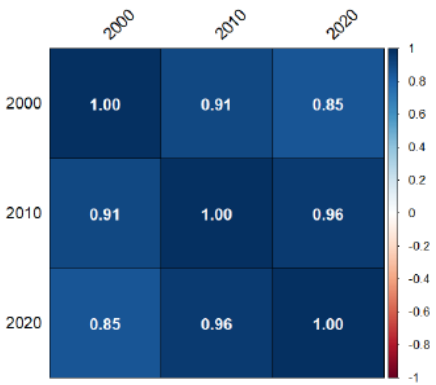
Hausmann and Hidalgo 2009 PNAS.

Extra homework for Sept 30

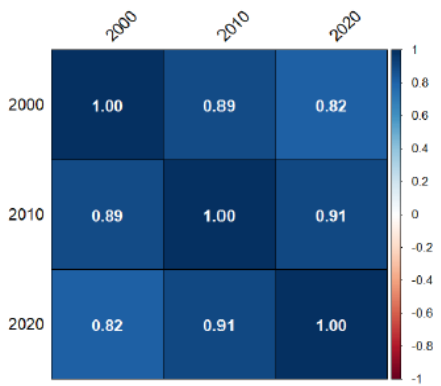
Illustrate the dynamics of complexity ranking in trade, technology and research.

1. Go to the country Complexity Rankings on oec.world and download the country data.
2. Investigate ranking dynamics in all dimensions across 2000-2010-2020. Compare correlations across time but within dimensions. Check cross-correlations across dimensions.
3. /Plus: Merge data with Economic Growth forcast tables https://oec.world/en/resources/methods/eci-growth-predictions. Calculate the coefficient of ECI indices using the Solow model of economic growth.
4. Prepare visualizations and discuss the results. Collect figures and text to a pdf (max 2 pages) and save it ”givenname_familyname_EC2026_HW1_extra.pdf”.
5. Include code and prompts in a zip file ”givenname_familyname_EC2026_HW1_documentation.zip”.
6. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
7. Extra homeworks covering points 1-2-4-5 can get 2 points. Point 3 gets plus 1 point.

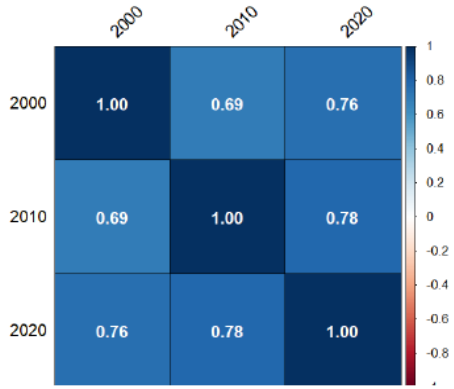
Coloring and caption



1. Figure: Spearman rank correlation matrix for Trade complexity across decades



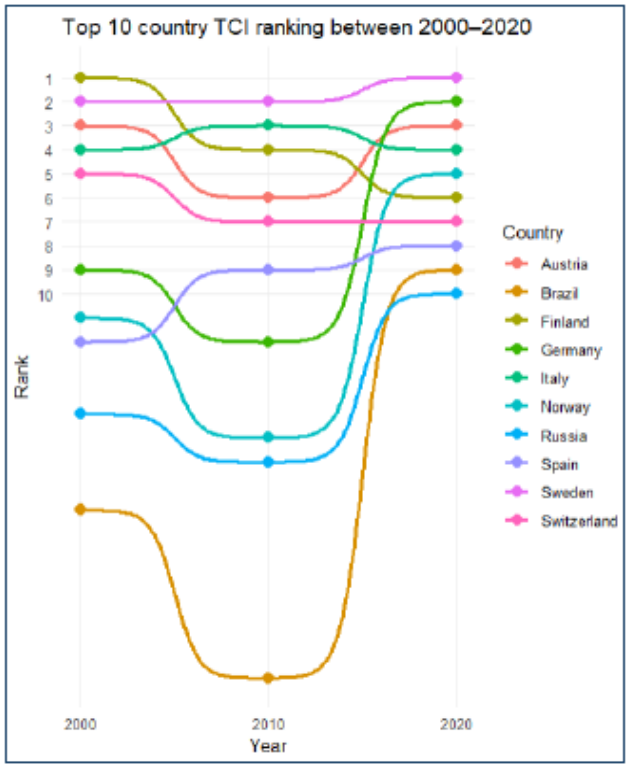
2. Figure: Spearman rank correlation matrix for Technology complexity across decades



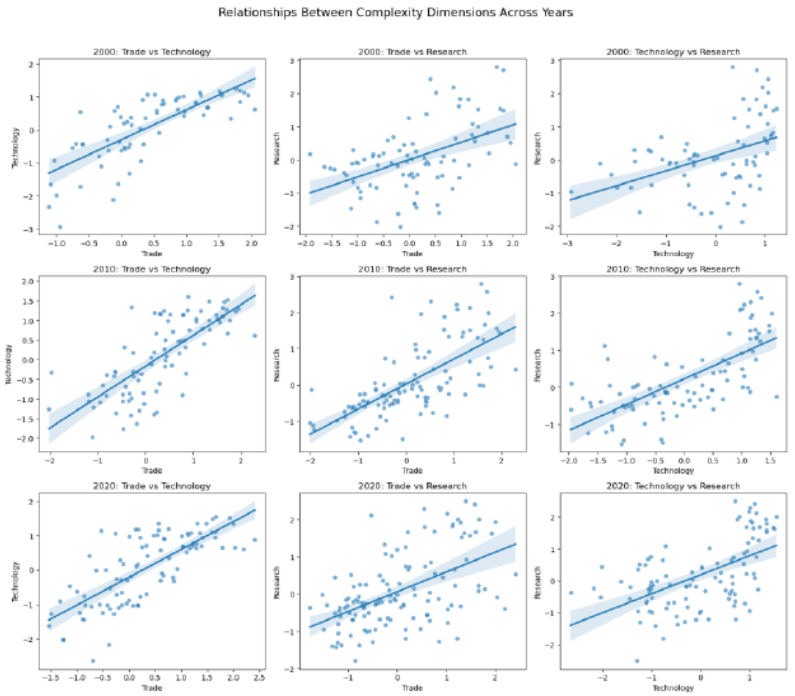
3. Figure: Spearman rank correlation matrix for Research complexity across decades

Period	2000-2010	2010-2020	2000-2020
Trade	0.936	0.940	0.839
Technology	0.900	0.907	0.826
Research	0.706	0.787	0.756

Font size



6. ábra: TCI ranking between 2000-2020



3. Figure Relationships Between Complexity Dimensions Across Years

Regression tables

- Necessary elements: coefficients, significance, constant, N, R2
- Correlation, multicollinearity: VIF tests, separate introduction of variables, interaction terms

	Model
(Intercept)	16.664***
	(0.729)
ln(GDP per capita)	-1.506***
	(0.073)
ECI Trade	0.587***
	(0.074)
ECI Tech	0.665***
	(0.075)
ECI Research	-0.089+
	(0.048)
Num. Obs.	86
R^2	0.854
R^2 Adj.	0.847

$p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Homework: correlations

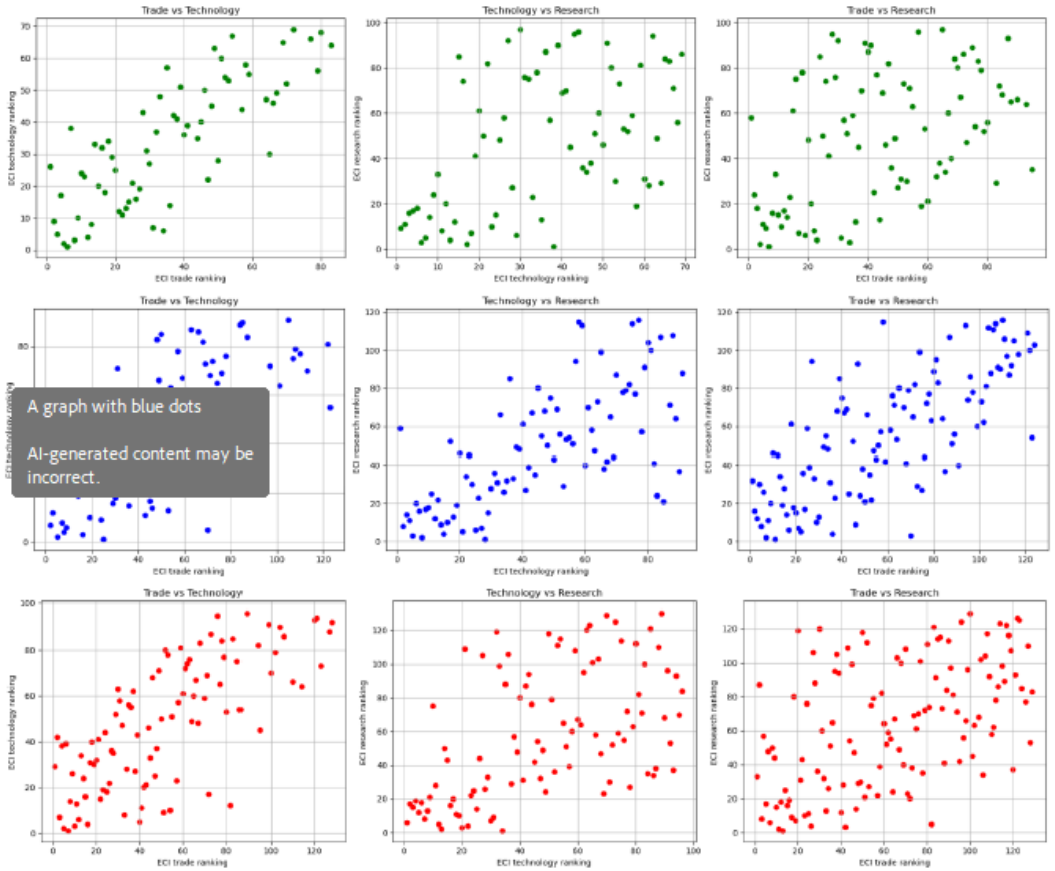


Figure 2: Figure 1: Comparison of ECI rankings between dimensions in 2000, 2010 and 2020

Homework: extras

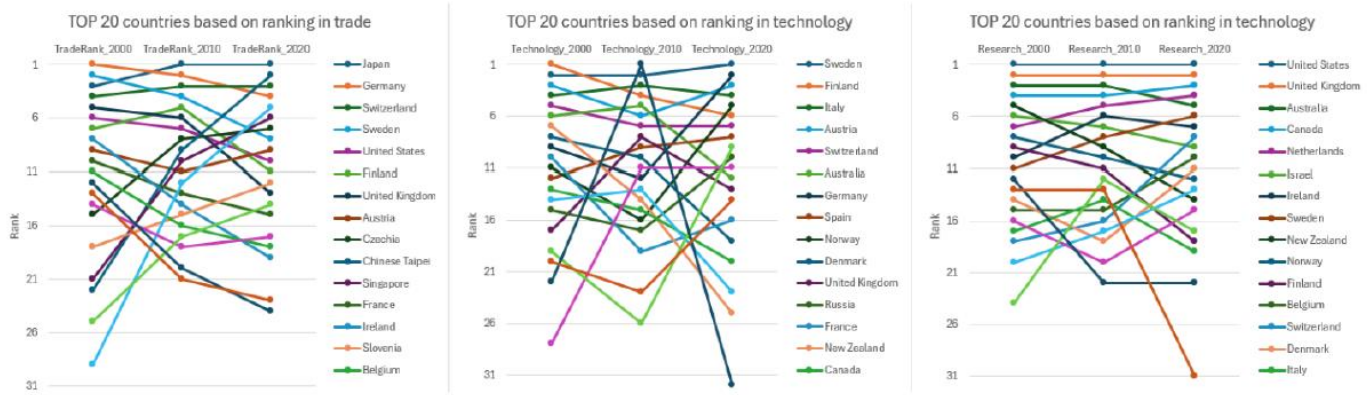
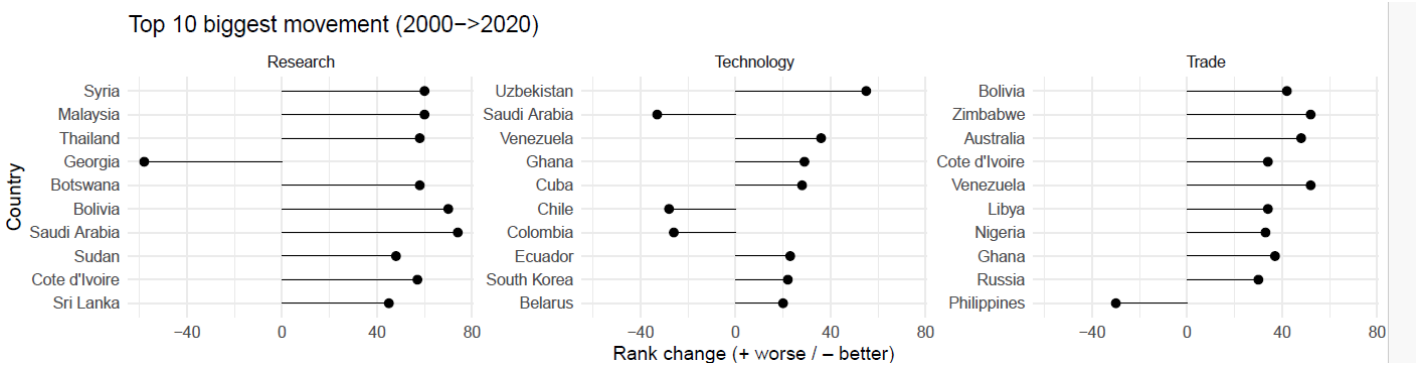


Figure 1: Top 20 countries based on the ranking of country complexity measure in 2000-2010-2020



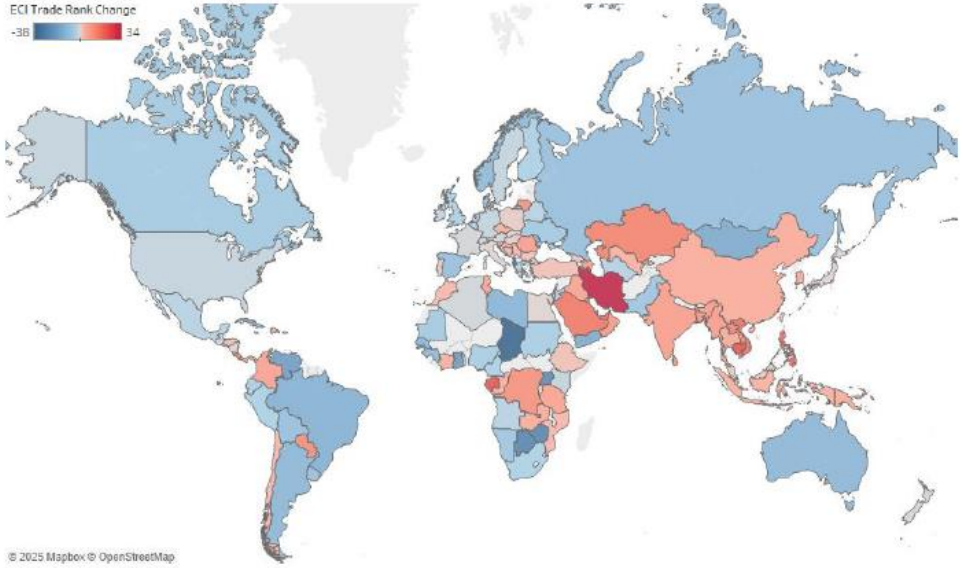


Figure 3: Changes in ECI Trade ranking from 2010 to 2020

Homework: extras

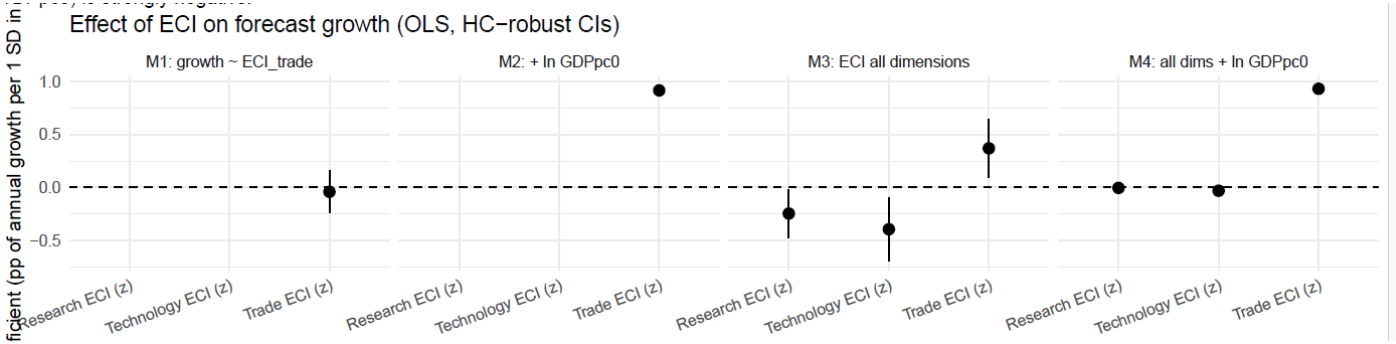
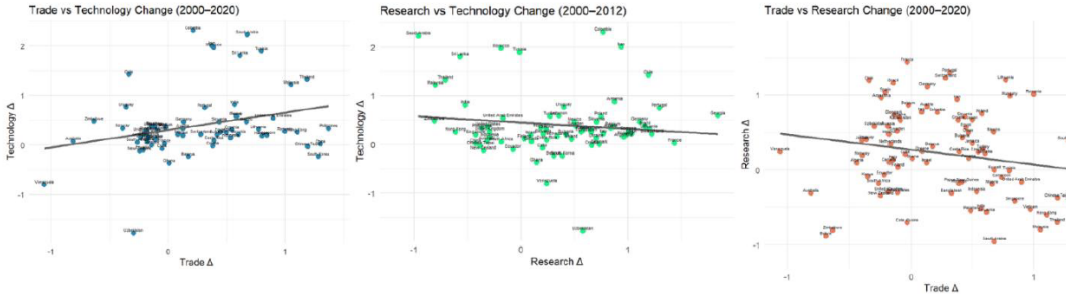


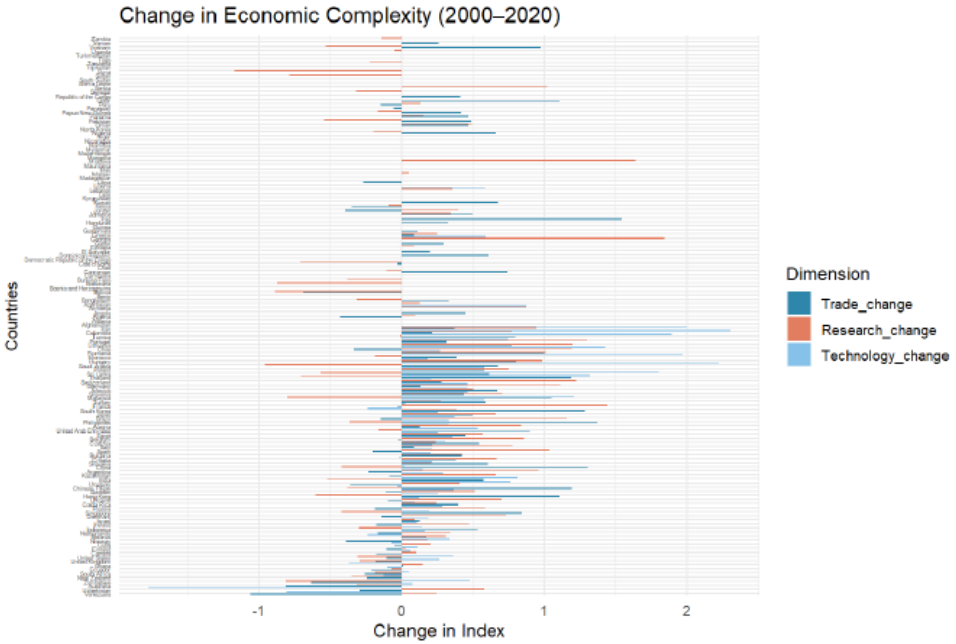


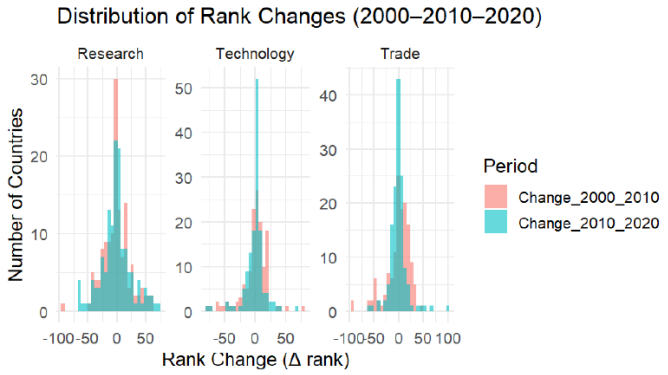
Figure 1: The relationship between the ECI in each category from 2000 to 2020

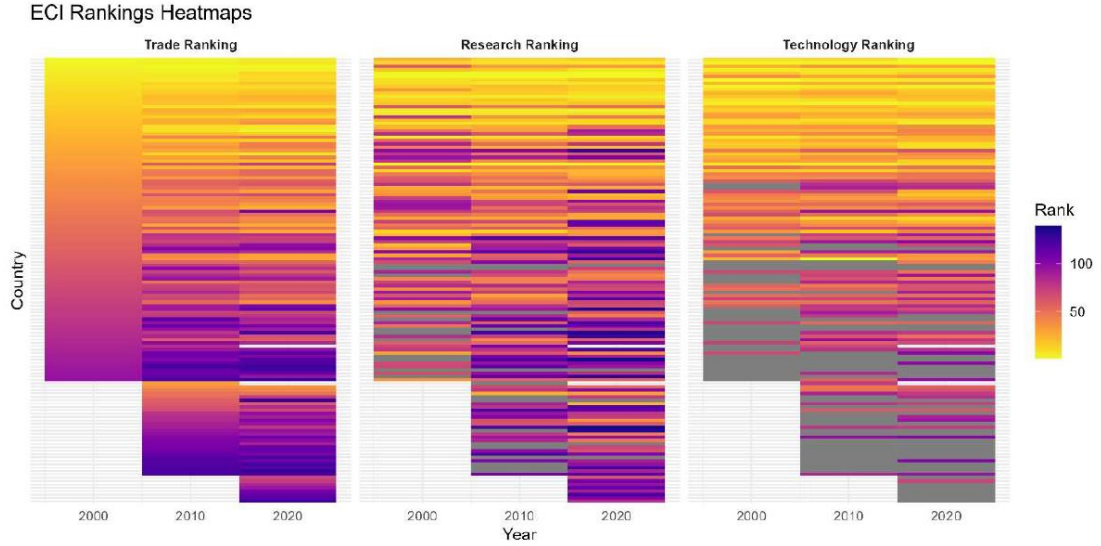
Homework: extras



The three scatter plots illustrate how trade, technology, and research complexity changed over time. Between 2000 and 2020, there is a mild positive relationship between trade and technology growth, meaning that countries with more complex trade structures often also advanced technologically. However, the links between research and the other two dimensions are weakly negative, indicating that increases in research complexity did not consistently correspond with progress in trade or technology.







Homework: estimation

Regression Analysis

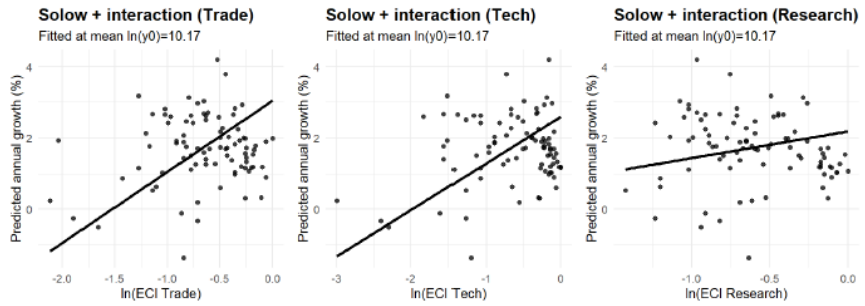


Figure 2

Using the Solow-style specification in Eq. (1), with predicted GDP per capita growth as the outcome and including $\ln(y_0)$, ECI, and their interaction, we find a consistent pattern across dimensions. The patterns can be seen in Figure 2.

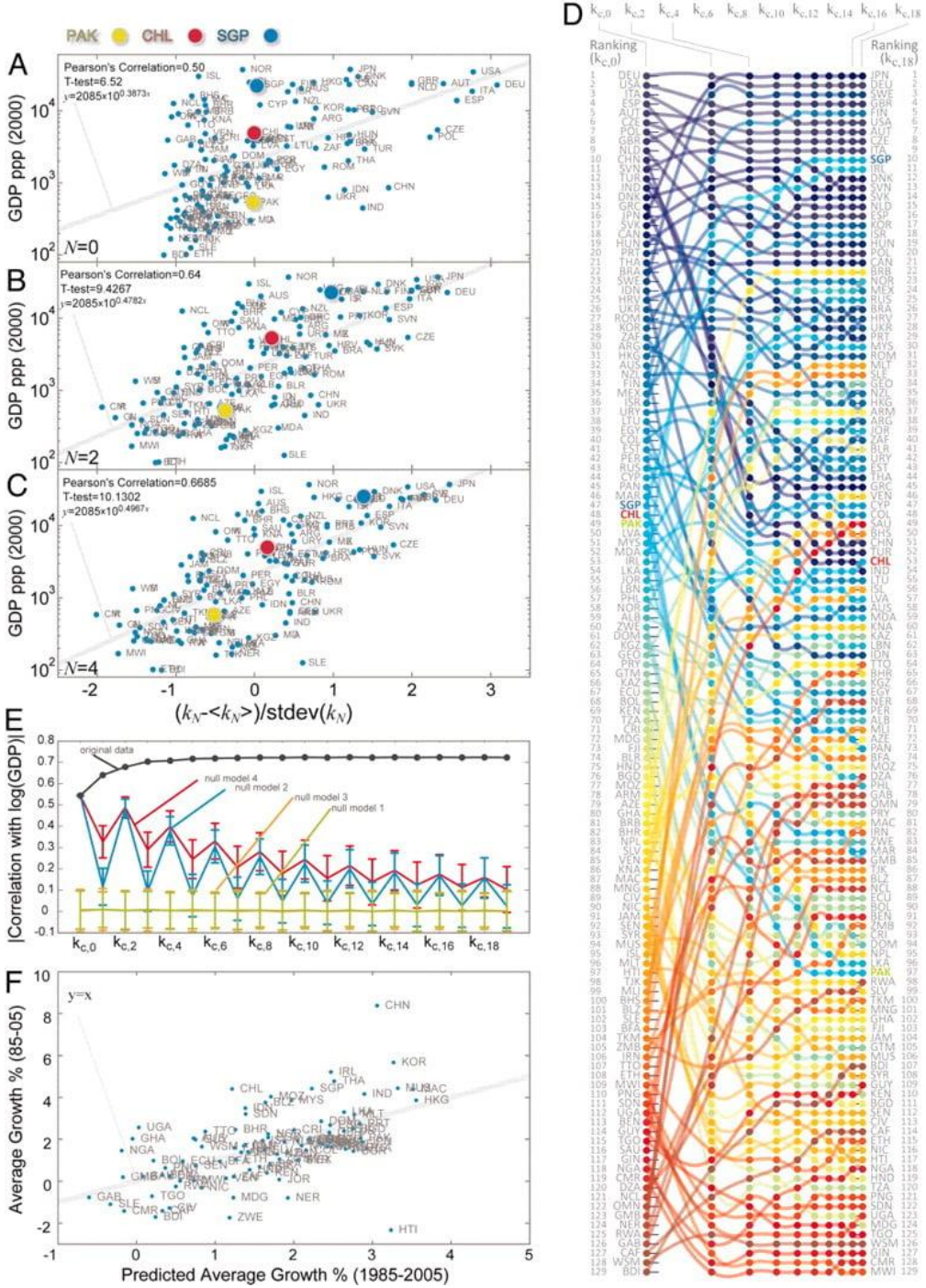
$$g_i = \beta_0 + \beta_1 \ln y_{0,i} + \beta_2 \ln(\mathrm{ECI}_i) + \beta_3 \ln y_{0,i} \ln(\mathrm{ECI}_i) + \varepsilon_i$$

Model estimates (Table 1) indicate that for trade and technology, the log-ECI main effect is positive and statistically significant at the 5% level of significance (Trade: $\hat{\beta}_2 \approx 4.63$, Technology: $\hat{\beta}_2 \approx 3.33$). In both models, the interaction term is negative and significant at the 10% level of significance (Trade: $\hat{\beta}_3 \approx -0.26$; Technology: $\hat{\beta}_3 \approx -0.20$), implying that the association between complexity and predicted growth attenuates as initial income rises. The convergence term is negative and highly significant in both cases ($\hat{\beta}_1 < 0$). By contrast, in the research model, neither the log-ECI main effect nor its interaction is statistically distinguishable from zero, and overall fit is markedly weaker (adjusted $R^2 \approx 0.35$). These patterns are consistent with the fitted lines: steep for Trade and Technology, comparatively flat for Research (though the slope does not matter here as it is statistically insignificant).

	Trade	Tech	Research
Intercept	18.86	16.85	12.38
ln(ECI)	4.62***	3.33***	4.04
ln(y0)	-1.56***	-1.40***	-1.00***
ln(y0)*ln(ECI)	-0.26*	-0.20*	-0.32
Adj. R^2	0.83	0.80	0.35
N	87	87	87

*Table 1 * $p<0.05$, ** $p<0.01$, *** $p<0.001$.*

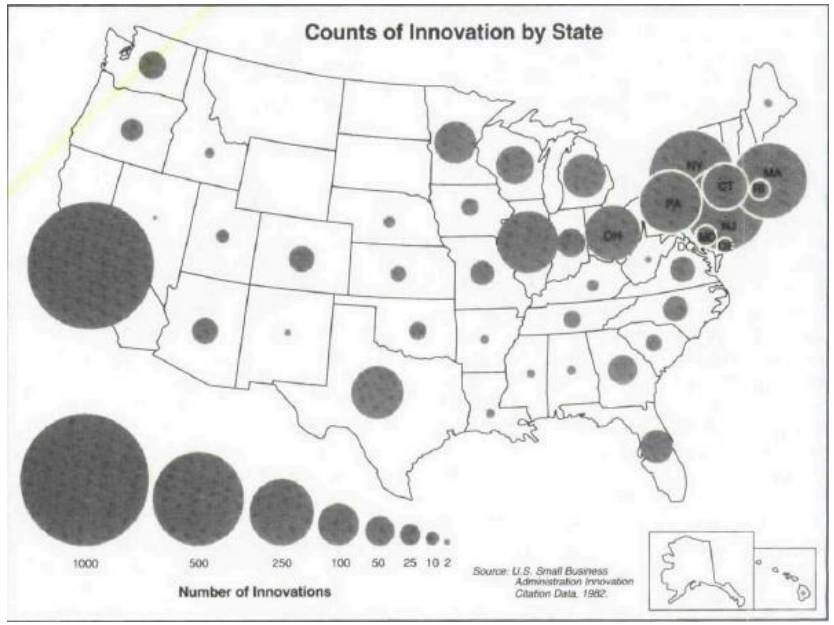
Quiz 2 – September 30



Reading for Oct 7

Audretsch and Feldman 1996

Quiz 3 – Oct 7 – Audretsch and Feldman AER 1996



Reading for Oct 14

Neffke, F., Otto, M., & Weyh, A. (2017). Inter-industry labor flows. Journal of Economic Behavior & Organization, 142, 275-292.

YOU CAN SKIP: GLAESER ET AL

HW 2 – for Oct 14

From a paper to a research niche.

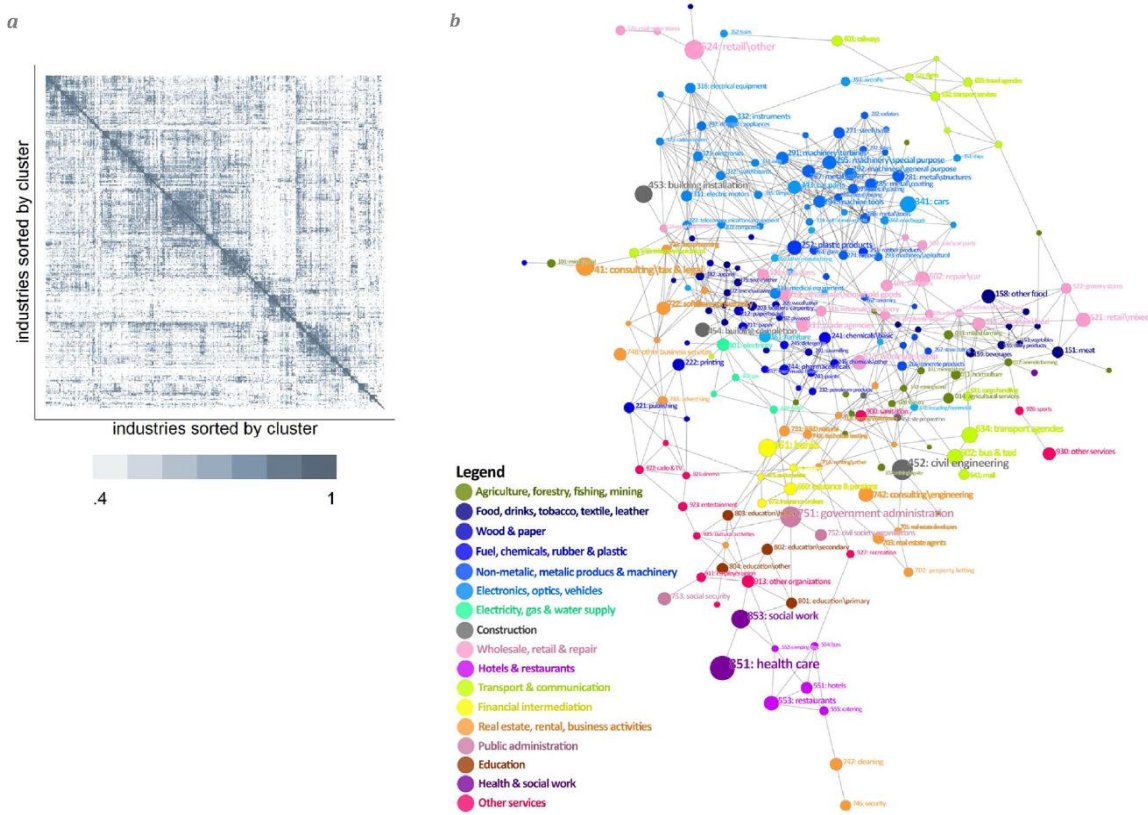
1. Reconstruct the niche of Neffke, Otto & Weyh (2017) in max. 5 sentences: What is the research problem? What did earlier relatedness measures (co-exports, co-occurrence, input-output) miss? What is the question and the contribution?
2. Find 2 journal articles published since 2020 that cite Neffke et al. (2017) (e.g., Google Scholar "Cited by"). For each, write 1–2 sentences on how they extended it: new **context, data, mechanism, method, or outcome**?
3. Formulate your own niche on skill-relatedness or labor flows (you can link it to your HW1 topic). Write a gap statement: **"Previous work has shown X, but we do not know Y. This matters because Z."** Then state one research question.
4. Name one data source that could answer your question, and who would use the answer (a company, a region, policy makers).
5. Max. 1 page pdf. Save the file: "givenname_familyname_EC2026_HW2.pdf". If you used LLMs, attach your prompt history.
6. Send it to the instructor by email. Deadline: October 13, Tuesday, 12noon. Homework sent after the deadline are not considered.
7. Homeworks can get max 3 points based on demonstration of understanding, creativity, clarity of communication. Students must be prepared for questions about their assignment asked in the class.

Extra homework for Oct 14

Relatedness networks from open data.

1. Choose an open dataset: country–product exports from oec.world (or the Atlas of Economic Complexity), or data published with a relatedness paper (replication files, supplementary materials, e.g., O*NET occupation–skill data as in Alabdulkareem et al. 2018).
2. Construct the relatedness network: either compute RCA from co-occurrence as in Hidalgo et al. 2007, or flow-based relatedness as in Neffke et al. 2017.
3. Visualize the network: Filter the edges, e.g., maximum spanning tree plus the strongest links. Color nodes by sector or category, size them by trade or employment, and add a legend and caption.
4. Describe the network: degree and weight distributions, clusters or communities, core vs. periphery.
5. **/Plus:** Compare two networks (e.g., two years, or trade vs. skill relatedness). Do strong links persist? Correlate edge weights.
6. Collect figures and text to a pdf (max 3 pages) and save it ”givenname_familyname_EC2026_HW2_extra.pdf”. Include code and prompts in a zip file ”givenname_familyname_EC2026_HW2_documentation.zip”.
7. Send it to the instructor by email. Deadline: October 13, Tuesday, 12noon. Homework sent after the deadline are not considered.
8. Extra homeworks covering points 1-2-3-4 can get 2 points. Point 5 gets plus 1 point.

Quiz 4 – Oct 14


a
industries sorted by cluster
industries sorted by cluster
.4
1
b
Legend
Agriculture, forestry, fishing, mining
Food, drinks, tobacco, textile, leather
Wood & paper
Fuel, chemicals, rubber & plastic
Non-metalic, metalic producs & machinery
Electronics, optics, vehicles
Electricity, gas & water supply
Construction
Wholesale, retail & repair
Hotels & restaurants
Transport & communication
Financial intermediation
Real estate, rental, business activities
Public administration
Education
Health & social work
Other services
851: health care
853: social work
751: government administration
452: civil engineering
742: consulting\engineering
453: building installation
341: cars
151: meat
158: other food
222: printing
634: transport agencies

Reading for Oct 20

Fleming, L., Charles King III., & Juda, A. I (2007). Small worlds of regional innovation. Organization Science, 18(6), 938-954.

HW 3 – for Nov 4

From niche to research pre-proposal – in pairs.

1. Form your pair for HW3 and the Research Proposal. Send both names to the instructor by October 27, Tuesday, 12noon.
2. **Use the comments you received on HW2.** You can build on your HW1 or HW2 topics, or choose a new one.
3. **Problem and gap:** build on at least 3 papers; at least one should study networks (e.g., Fleming et al. 2007, Abbasiharofteh et al. 2023). End with a gap statement.
4. **Research question and hypothesis:** explain the mechanism – why does the network matter?
5. **Data and method:** what are the nodes and links, which data source, which network measure (e.g., clustering, path length, segregation, robustness)? Are the data accessible? What is your plan B?
6. **Expected results and impact:** what do you expect to find if your hypothesis holds? Describe the impact in as much detail as possible: why is the exercise you propose important, what do you want to improve, and for whom?
7. Max. 2 pages pdf including references, Times New Roman, font size 10–12. You can add one figure on a third page. Save the file: ”familyname1_familyname2_EC2026_HW3.pdf”, and attach your prompt history. Send it to the instructor and TA by email. Deadline: November 3, Tuesday, 12noon. Homework sent after the deadline are not considered.
8. Each student can get 3 points. Points are deducted for conceptual errors, unclear communication, and data problems.

Extra homework for Nov 4

Replicate a published network study (individually – NOT IN PAIRS).

1. Download the town-level network of the iWiW online social network from Lengyel et al. (2015, PLoS ONE) on Dryad: doi:10.5061/dryad.33ps4. Read the paper's data and methods sections first.
2. **Distance decay:** compute the probability of a tie, P(d) = L(d)/N(d), in 5 km distance bins and estimate the exponent on a log–log plot (paper: ≈ −0.6). Add town coordinates from public sources if needed.
3. **Communities:** run Louvain on the network weighted by tie counts (w1) and by observed relative to expected ties (w2). Map the communities and compare them with the paper (5 and 14 modules).
4. Discuss which results you could reproduce, where your numbers differ, and why.
5. **/Plus:** extend the analysis, e.g., test whether county borders or the Danube reduce ties after controlling for distance.
6. Collect figures and text to a pdf (max 3 pages) and save it "familyname_EC2026_HW3_extra.pdf". Include code and prompts in a zip file "familyname_EC2026_HW3_documentation.zip". Send it to the instructor by email. Deadline: November 3, Tuesday, 8pm. Homework sent after the deadline are not considered.
7. Extra homeworks covering points 1-2-3-4 can get 2 points. Point 5 gets plus 1 point.

Reading for Nov 4

Tóth, G., Wachs, J., Di Clemente, R., Jakobi, Á., Ságvári, B., Kertész, J., & Lengyel, B. (2021). Inequality is rising where social network segregation interacts with urban topology. Nature Communications, 12(1), 1143.

nature COMMUNICATIONS

ARTICLE

https://doi.org/10.1038/s41467-021-21465-0 OPEN

Inequality is rising where social network segregation interacts with urban topology

Gergő Tóth[1,2,13], Johannes Wachs[3,4,13], Riccardo Di Clemente[5,6], Ákos Jakobi[7,8], Bence Ságvári[1,9,10], János Kertész[11] & Balázs Lengyel[1,10,12]✉

Reading for Nov 11

Lengyel, B., Bokányi, E., Di Clemente, R., Kertész, J., & González, M. C. (2020). The role of geography in the complex diffusion of innovations. *Scientific Reports*, *10*(1), 15065.